

24281149

5-1. 逻辑功能:
组合电路: 无记忆性, 电路在任意时刻的输出状态, 仅取决于该时刻的输入信号.

时序电路: 有记忆性. 不仅取决于该时刻的输入信号, 还与电路原来的状态有关.

电路结构上:

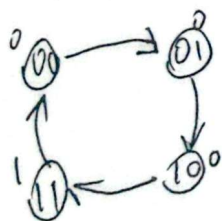
组合电路: 仅由逻辑门组成.

时序电路: 由组合逻辑电路和存储电路组成.

5-4: 驱动方程: $D_0 = \overline{Q_0}$
 $D_1 = Q_1 \oplus Q_0$

状态方程: $Q_0^{n+1} = \overline{Q_0^n}$
 $Q_1^{n+1} = Q_1^n \oplus Q_0^n$

输出方程: $F = Q_1^n Q_0^n$



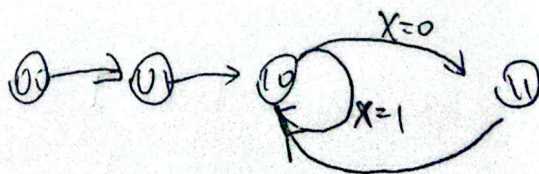
功能: 同步模4二进制加法计数器

5-5. $J_1 = K_1 = Q_0 \overline{Q_1}$
 $J_0 = \overline{X} Q_1, K_0 = 1$

$$Q_1^{n+1} = (Q_0^n \overline{Q_1^n}) \oplus Q_1^n = Q_1^n + Q_0^n$$

$$Q_0^{n+1} = J_0 \overline{Q_0^n} + \overline{K_0} Q_0^n = (\overline{X} Q_1^n) \overline{Q_0^n} + Q_0^n$$

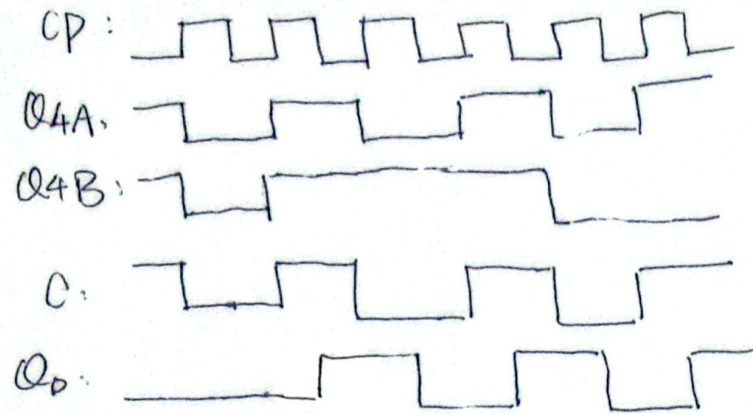
$$Y_1 = Q_1, Y_0 = Q_0$$



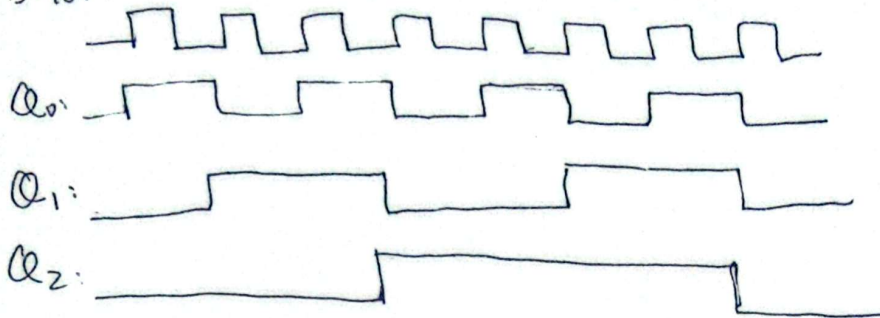
功能: 自动计数到2, 然后受X控制
 $X=1$ 则保持2
 $X=0$ 则在2和3间循环



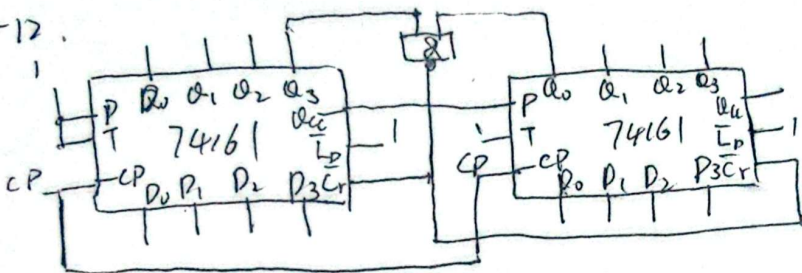
5-7:



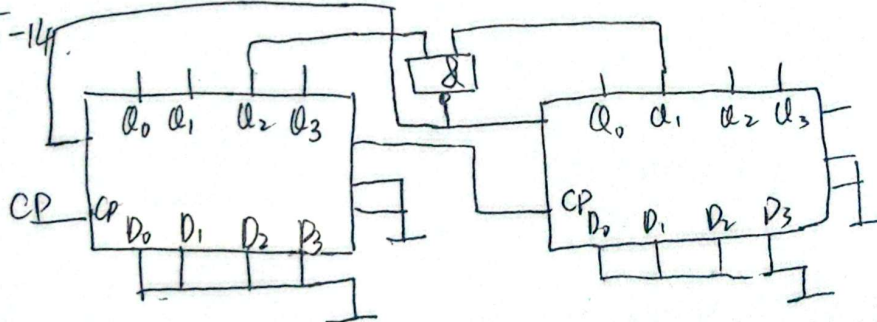
5-10:



5-12:



5-14:



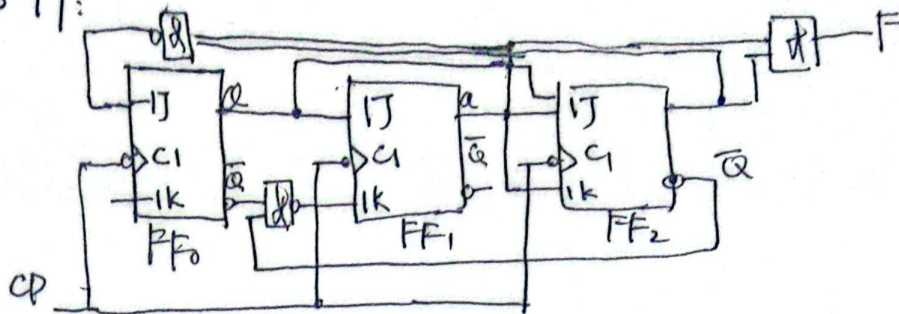
5-16: 当 $I/C=1$ 时, 因为 $\overline{I_0}=1, \overline{C_r}=1$, 所以 74161 为二进制加法计数器
当 $I/C=0$ 时, $\overline{I_0}=0, \overline{C_r}=1$, 74161 实现移位功能

使 $Q_3^{n+1} Q_2^{n+1} Q_1^{n+1} Q_0^{n+1} = Q_2^n Q_1^n Q_0^n D$ 。 $Q_3^{n+1} = Q_2^n, Q_2^{n+1} = Q_1^n, Q_1^{n+1} = Q_0^n, Q_0^{n+1} = D$

故 $I/C=1$ 时为计数器, $I/C=0$ 时为移位寄存器。



5-19:



5-22

